Statistical Estimation of Canteen Arrival Overload Probability Using Poisson Distribution

Faraz Akhtar Khan
School of Computer Science Engineering and Information Systems
Vellore Institute of Technology
Vellore, India

Mayur Krishna
School of Computer Science Engineering and Information Systems
Vellore Institute of Technology
Vellore, India

Malhar Bhosale
School of Computer Science Engineering and Information Systems
Vellore Institute of Technology
Vellore, India

Manimaran A
School of Advanced Sciences
Vellore Institute of Technology
Vellore, India

***Abstract*—University canteens see sharp, short-lived rises in arrivals during the lunch peak, and the practical question for students and managers is how likely a given time slot is to exceed what the serving counters can handle. This paper models student arrivals in each 15-minute slot as a Poisson count, estimates a separate rate λ_t for every slot with an exact confidence interval, tests the Poisson assumption slot by slot, and converts the tail probability $P(X_t > C)$ into low, medium and high overload-risk labels. The analysis is then extended from description to prediction: a multiple linear regression on log counts gives day-specific rates for each day of the week; a Poisson–Gamma mixture tests for extra day-to-day variation; Monte Carlo simulation gives the probability that each slot is the day's peak; and Bayes' theorem with a Gamma prior provides a same-day update. Every forecast is checked on days the model has not seen with a rolling one-day-ahead backtest. On 40 days of recorded lunch-time arrivals, the riskiest slot is 12:45–13:00 ($P = 55.9\%$ at $C = 30$), 13:00–13:15 is significantly quieter, weekends bring about 26% fewer arrivals, and on a regular Monday the peak falls in 12:30–13:00 with probability 83%. On 20 unseen days the busiest slot was named correctly 60% of the time against 40% by chance and the overload forecasts had a Brier skill of 31% over a per-slot average, while the gain in point accuracy was not statistically significant. All methods use standard probability and statistics, and the implementation is released as an interactive dashboard.**

***Index Terms*—Poisson distribution, arrival process, overload probability, peak-time forecasting, Bayesian updating, negative binomial, hypothesis testing, university canteen.**

I. Introduction

University canteens experience changing arrival intensity during lunch. When more students arrive in a short interval than the counters can serve, queues form, the dining area becomes crowded and popular dishes run out. Decisions about when to eat, how many counters to open and when to stagger breaks are usually taken by intuition rather than from measured arrival behaviour.

Existing canteen studies treat the canteen as a complete system. Tang *et al.* [1] collected camera data at a university canteen and built a pedestrian-flow model of the whole meal process, and a cellular-automaton model followed [2]. Queueing studies set the number of serving windows from assumed Poisson arrivals [3]–[5]. These models are valuable but heavy, and they leave a simple question unanswered: *what is the probability that arrivals exceed a chosen level during a particular time slot, and when will today's peak occur?*

This paper answers that question with standard probability and statistics. Its contributions are:

1) A per-slot Poisson model of canteen arrivals with exact confidence intervals, slot-by-slot goodness-of-fit and dispersion tests, and an overload probability $P(X_t > C)$ mapped to risk labels.
2) A day-specific forecast of slot rates by multiple linear regression on day-of-week and day-type indicators, with a Poisson–Gamma model for day-to-day variation.
3) The probability that each slot is the day's peak, obtained by simulation, and a Bayesian update of the rest-of-day forecast from the counts observed so far.
4) A validation protocol that forecasts each day from earlier days only and tests the forecasts with proportion, paired, sign, Kolmogorov–Smirnov and chi-square tests, and the Brier score.

The methods are applied to 40 days of recorded arrival counts at a university canteen (Section III).

The rest of the paper is organised as follows. Section II reviews related work and the gaps it leaves. Section III describes the data and Section IV the methodology. Section V presents results, Section VI discusses them and their limitations, and Section VII concludes.

II. Related Work

A. Canteen flow and queueing

Tang *et al.* [1] counted arrivals and departures in 5-minute segments at a Beihang University canteen between 10:30 and 13:30 and calibrated a model of window choice, movement, congestion delay and meal duration. They note that arrivals follow a Poisson distribution within short periods with a rate that changes every few minutes, and that days differ with the

class timetable; the arrival rate, however, is an input to their simulation rather than a quantity that is tested or predicted. Queueing studies set windows, seats and opening times with M/M/n/m and M/M/C models [3], [4]; Jiao *et al.* [5] simulate one constant arrival rate of 19.3 students per minute and find seven windows optimal. In all of these the Poisson arrival assumption is an input, not a finding.

B. Testing Poisson arrivals

Brown *et al.* [6] established a statistical treatment of service arrivals and a test that a point process is Poisson. Kim and Whitt [7] show that nonhomogeneous Poisson models fit call-centre and hospital arrivals only after careful analysis: long subintervals and pooled days create overdispersion. Green *et al.* [8] show that time-varying demand needs short, piecewise-constant rates, and Leemis [9] estimates such rates from repeated days. Cameron and Trivedi [10] give tests for overdispersion in Poisson models.

C. Forecasting arrival rates

Weinberg *et al.* [11] forecast within-day call-centre rates as a daily level times a weekday shape and update the forecast during the day; Shen and Huang [12] forecast quarter-hour Poisson rate profiles and report that intraday updating cut staffing error by 14.5%. Aldor-Noiman *et al.* [13] add special-event indicators and compare against an "industry model" that averages past slots, which is the per-slot average used by descriptive canteen analyses. Matteson *et al.* [14] stress that small hourly counts should be modelled as Poisson directly rather than through normal approximations. Ibrahim *et al.* [15] survey the field: arrivals are overdispersed and correlated within a day, which a Gamma "busyness factor" captures, and simple fixed-effect models are hard to beat.

D. Campus dining and forecast verification

Shah *et al.* [16] forecast daily customer totals in a university self-service restaurant with machine learning, although their own data show strong hourly peaks; Variskhanov and Arreeras [17] issue a day-level "early peak" badge from Wi-Fi telemetry. For probability forecasts of a yes/no event, the Brier score [18] and a skill score against a reference forecast [19] are the standard checks.

E. Research gaps

Table I summarises what the reviewed work leaves open. Four gaps motivate this paper: (G1) no per-slot overload probability for a canteen; (G2) the Poisson assumption is built in rather than tested; (G3) no prediction of a specific day's rates or of the peak time, with or without the day's early counts; and (G4) no test of canteen forecasts on unseen days. These are gaps of focus and application, not a claim that Poisson modelling of arrivals is new.

TABLE I
REPRESENTATIVE STUDIES AND WHAT THEY LEAVE OPEN

Study	Approach	Left open for this work
Tang *et al.* [1]	Pedestrian-flow model of a canteen	Arrival rate is an input; no slot overload probability
Jiao *et al.* [5]	M/M/C simulation, one rate	Poisson assumed; rate constant all lunch
Kim & Whitt [7]	Tests of Poisson arrivals	Call centres and hospitals, not canteens
Weinberg *et al.* [11], Shen & Huang [12]	Rate forecasting, intraday update	Large-count approximations; staffing, not overload
Shah *et al.* [16]	ML forecast of campus dining	Daily totals only; no probabilities

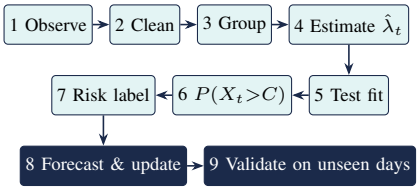


Fig. 1. Statistical framework. Light boxes: estimation and checking; dark boxes: prediction and validation.

III. DATA

Recorded arrivals. Student arrivals at the canteen were recorded in fixed 15-minute slots covering the start of the lunch period, 12:30–13:15 (three slots), on 40 consecutive days from 5 January to 13 February 2026. The record includes weekends: 30 weekdays and 10 Saturdays and Sundays, marked with the day type "Weekend". Each record holds the date, the weekday, the day type, the slot and the number of students who entered in that slot, in the style of the interval counts of [1].

Cleaning. Missing, negative, fractional, unreadable and duplicate counts are dropped rather than imputed, because imputing the mean would shrink the variance and bias the dispersion test. All 120 records passed these checks, giving 120 slot observations (40 per slot). Counts range from 17 to 41 students per slot, with a mean of 30.0.

IV. METHODOLOGY

Fig. 1 shows the nine-step framework. Steps 1–7 estimate and check overload risk; steps 8–9 forecast and validate.

A. Poisson model and estimation

Let X_t be the number of students arriving in slot t. The model assumes

$$X_t \sim \text{Poisson}(\lambda_t), \qquad P(X_t = x) = \frac{e^{-\lambda_t}\lambda_t^x}{x!}, \tag{1}$$

$x = 0, 1, 2, \ldots$, with $E[X_t] = \text{Var}(X_t) = \lambda_t$. For slot t observed on m days with counts $x_1, \ldots, x_m$, the maximum-likelihood estimate is the sample mean $\hat{\lambda}_t = \sum x_i/m$. Because $S = \sum x_i \sim \text{Poisson}(m\lambda_t)$, the exact (Garwood) interval is [20]

$$\left[\frac{\chi^2_{\alpha/2}(2S)}{2m}, \frac{\chi^2_{1-\alpha/2}(2S+2)}{2m}\right]. \tag{2}$$

B. Checking the Poisson assumption

Two tests are applied to every slot at $\alpha = 0.05$. The index-of-dispersion test uses $D = (m-1)s^2/\bar{x} \sim \chi^2_{m-1}$ under H_0, two-sided. The chi-square goodness-of-fit test compares observed and expected frequencies, $\chi^2 = \sum_i (O_i - E_i)^2/E_i$ with $k-2$ degrees of freedom, after merging adjacent counts until every expected frequency is at least 5 [21]. Because K slots are tested, the number of rejections is compared with a Binomial(K, α) distribution.

C. Overload probability and risk

With capacity C per slot (threshold $K = C + 1$),

$$P(X_t > C) = 1 - \sum_{x=0}^{C} \frac{e^{-\hat{\lambda}_t}\hat{\lambda}_t^x}{x!}. \tag{3}$$

A slot is labelled low risk below 20%, high risk above 50% and medium otherwise; these thresholds are project decision rules. The probability is also evaluated at both ends of the confidence interval for λ_t, and the smallest capacity keeping it below a target is reported for planning. To check that the quietest slot is genuinely quieter than the runner-up, the exact conditional test $S_1 \mid (S_1 + S_2) \sim \text{Binomial}(S_1 + S_2, m_1/(m_1 + m_2))$ is used [22].

D. Day-specific rates by regression

To predict a particular day, the slot rate is allowed to depend on the day of the week d and the day type e. Multiple linear regression with dummy variables is fitted by least squares to log counts:

$$\ln(x_{dt} + 0.5) = \beta_0 + \alpha_t + \gamma_d + \delta_e + \varepsilon_{dt}, \tag{4}$$

with 12:30 and Monday as reference levels; a day type that coincides with whole days of the week (here the weekend, which is exactly Saturday and Sunday) is captured by those day terms and gets no separate coefficient. Each coefficient is tested with $t_j = \hat{\beta}_j/\text{SE}(\hat{\beta}_j) \sim t_{n-p}$, and fitted values are returned to the count scale with Duan's smearing factor $\overline{e^{\hat{\varepsilon}}}$ [23]. Weekday effects on daily totals are also tested with one-way ANOVA and the Kruskal–Wallis test, and weekends against weekdays with Welch's t-test and the Mann–Whitney U test [24].

E. Day-to-day variation: Poisson–Gamma model

Whole days can be busier or quieter than predicted [15]. Let each day carry a busyness factor $W \sim \text{Gamma}(1/v, 1/v)$ with mean 1 and variance v. By the theorem of total probability,

$$P(X_t = x) = \int_0^\infty \frac{e^{-w\lambda_t}(w\lambda_t)^x}{x!}\, g(w)\, dw, \tag{5}$$

a negative binomial with $E[X_t] = \lambda_t$ and $\text{Var}(X_t) = \lambda_t + v\lambda_t^2$. The Pearson statistic $\chi^2_P = \sum (x - \hat{\lambda})^2/\hat{\lambda}$ follows χ^2_{n-p} under the plain Poisson model, and its excess gives the moment estimate $\hat{v} = (\chi^2_P - (n-p))/\sum \hat{\lambda}$ [10]. With $v = 0$ the model reduces to the Poisson model above.

F. Peak-time probability

For a chosen day, $N = 20{,}000$ lunch periods are simulated: draw W, then $X_t \sim \text{Poisson}(W\hat{\lambda}_t)$ independently for each slot. The share of simulated days on which slot t has the largest count (ties broken at random) estimates $P(\text{slot } t \text{ is the peak})$; the same draws give $P(\text{at least one slot exceeds } C)$.

G. Same-day Bayesian update

After the first k slots of today have been counted, Bayes' theorem with the Gamma prior $W \sim \text{Gamma}(a, a)$, $a = 1/v$, gives the conjugate posterior

$$W \mid x_1, \ldots, x_k \sim \text{Gamma}\Big(a + \sum_{t \le k} x_t,\ a + \sum_{t \le k} \hat{\lambda}_t\Big), \tag{6}$$

and each later slot then has a negative binomial predictive distribution, which yields the updated $P(X_t > C)$.

H. Validation on unseen days

The first half of the record (20 days) is used for training; each later day is forecast from all earlier days only and compared with what happened. The model is compared with the per-slot average used in descriptive analyses (the "industry model" [13]). The checks are: coverage of 95% prediction intervals (z-test of a proportion); whether the predicted busiest slot is correct more often than chance (one-sided z-test); daily mean absolute error against the baseline (paired t-test and sign test); calibration of the full forecast distribution through randomised probability integral transform (PIT) values, $u = F(x-1) + V[F(x) - F(x-1)]$ with $V \sim U(0,1)$, tested for uniformity by Kolmogorov–Smirnov and chi-square tests; and the Brier score $B = \frac{1}{N}\sum (p_i - o_i)^2$ of the overload forecasts with skill $1 - B_{\text{model}}/B_{\text{baseline}}$ [18], [19]. Spearman rank correlation between consecutive slots and a Wald–Wolfowitz runs test on daily residual signs check within-day and day-to-day dependence.

I. Implementation

The methods are implemented in Python with NumPy, pandas and SciPy, covered by 72 automated tests, and presented in an interactive Streamlit dashboard whose home page answers "when should I go?" in plain language. All figures and tables in this paper are generated from the same code.

V. Results

All results use the 40 recorded days, capacity $C = 30$ per slot and $\alpha = 0.05$.

A. Arrival rates and model fit

The arrival rate is 30.5 at 12:30–12:45, peaks at 31.5 at 12:45–13:00 and falls to 27.9 at 13:00–13:15 (Fig. 2, Table II). The confidence intervals of the first two slots straddle the capacity line, so an average day is at or just above capacity there. The chi-square test rejects the Poisson model in none of the three slots, against 0.15 rejections expected by chance; for the peak slot, $\chi^2 = 0.98$ on 5 degrees of freedom, $p = 0.964$ (Fig. 3). The quietest slot, 13:00–13:15 ($\hat{\lambda} = 27.9$), is significantly quieter than the runner-up 12:30–12:45 ($\hat{\lambda} = 30.5$; exact test $p = 0.015$).

TABLE II
PER-SLOT ESTIMATES, MODEL CHECKS AND OVERLOAD PROBABILITIES ($C = 30$, 40 DAYS)

Slot	$\hat{\lambda}_t$	95% CI	Var/Mean	χ^2 GOF p	$P(X_t > C)$ Poisson, %	$P(X_t > C)$ Poisson–Gamma, %	Risk
12:30–12:45	30.52	[28.84, 32.29]	0.84	0.890	48.97	48.97	Med.
12:45–13:00	31.50	[29.78, 33.29]	0.88	0.964	55.93	55.93	High
13:00–13:15	27.88	[26.26, 29.56]	1.25	0.331	30.12	30.12	Med.

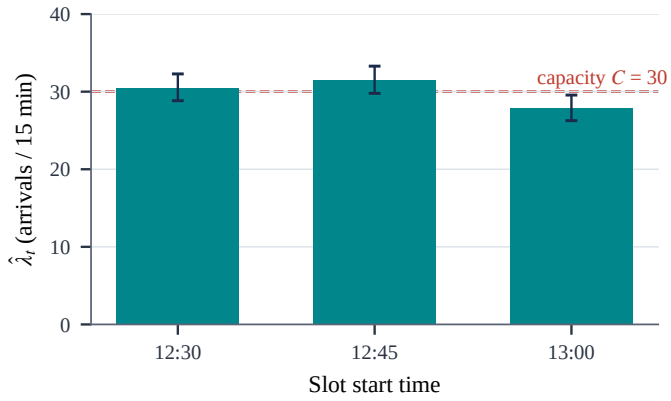


Fig. 2. Estimated arrival rate per slot with exact 95% confidence intervals and the capacity $C = 30$.

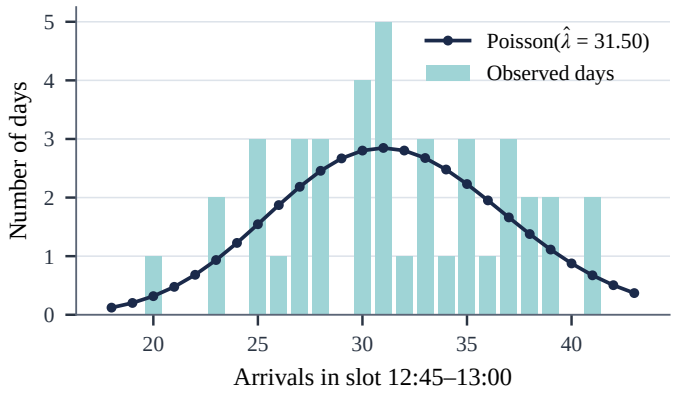


Fig. 3. Observed distribution of daily counts in the peak slot against the fitted Poisson model.

B. Overload risk

One slot, 12:45–13:00, is high risk ($P = 55.9\%$); 12:30–12:45 (49.0%) and 13:00–13:15 (30.1%) are medium risk, and no slot is low risk (Fig. 4). Keeping $P(X_t > C)$ below 20% in every slot would need a capacity of 36 per slot at the peak, against 30 at present. Once slot and day effects are removed, the Pearson test finds no extra day-to-day variation ($\chi^2_P = 59.7$ on 111 df; $\hat{v} = 0$): the counts vary, if anything, less than a Poisson model allows (dispersion ratio 0.54). The Poisson–Gamma model therefore reduces to the Poisson model and gives the same overload probabilities.

C. Day-specific forecast and peak time

The regression explains 55.5% of the variance of log counts (adjusted $R^2 = 0.523$). Relative to Monday, Saturday

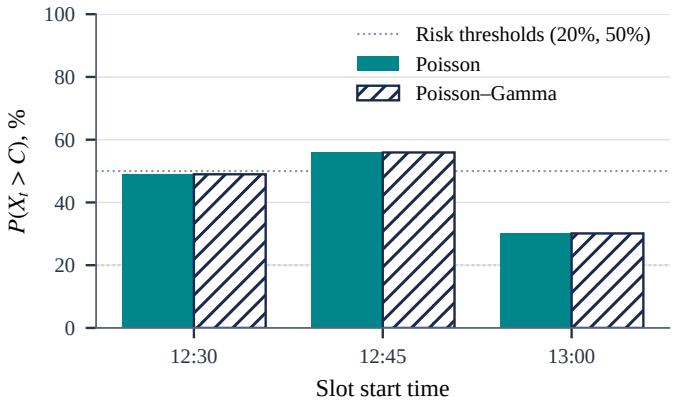


Fig. 4. Overload probability per slot with the risk thresholds. The Poisson–Gamma bars equal the Poisson bars because no extra day-to-day variation was found ($\hat{v} = 0$).

multiplies expected arrivals by 0.724 ($t = -6.87$, $p < 0.001$) and Sunday by 0.745 ($t = -6.25$, $p < 0.001$); the 13:00 slot is 9.3% quieter than 12:30 ($p = 0.002$), and Tuesday to Friday do not differ significantly from Monday (Table III). On daily totals the weekdays do not differ (ANOVA $p = 0.371$, Kruskal–Wallis $p = 0.414$), whereas weekends are clearly quieter (71.0 against 96.2 arrivals over the three slots; Welch $t = -10.21$, $p < 0.001$; Mann–Whitney $p < 0.001$).

Fig. 5(a) shows the resulting forecasts. On a regular Monday the probability that at least one slot is overloaded is 94.9%, with 1.8 overloaded slots expected; on a Friday 97.4% and 2.1; on a Saturday only 21.9% and 0.2. The peak is most likely at 12:45–13:00 (46%), with 12:30–12:45 close behind, so the peak falls in 12:30–13:00 with probability 83% (Fig. 5(b)). Neighbouring slots have similar rates, so the peak time is genuinely uncertain on any single day even when the rates are known.

D. Same-day update

Because $\hat{v} = 0$, the Gamma prior on the day's busyness factor is concentrated at 1, and the counts in the first slot carry no information about the later slots beyond what the regression already predicts: a first slot 15% busier than forecast leaves the 12:45–13:00 overload probability at 72.4%. On this canteen, a busy start therefore does not signal a busy rest of lunch; the update would matter only where whole days vary more than Poisson.

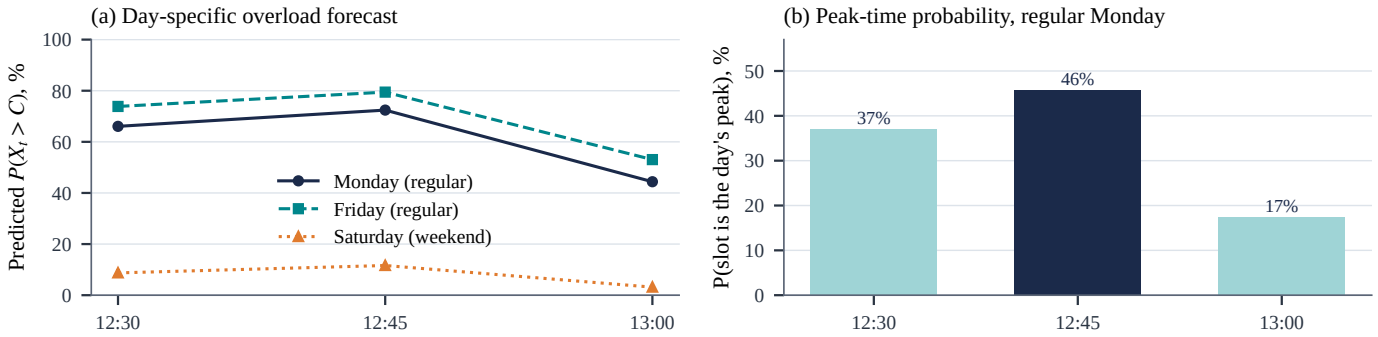


Fig. 5. (a) Predicted overload probability for a Monday, a Friday and a Saturday. (b) Probability that each slot is the busiest of the day on a regular Monday, from 20,000 simulated lunch periods.

TABLE III
REGRESSION OF $\ln(x + 0.5)$ ON SLOT, WEEKDAY AND DAY TYPE

Term	Estimate	SE	t	p	$e^{\hat{\beta}}$
Intercept†	3.5037	0.0362	96.79	<0.001	33.238
Slot 12:45	0.0313	0.0301	1.04	0.301	1.032
Slot 13:00	−0.0978	0.0301	−3.25	0.002	0.907
Tuesday	−0.0621	0.0449	−1.38	0.169	0.940
Wednesday	0.0035	0.0449	0.08	0.939	1.003
Thursday	−0.0179	0.0449	−0.40	0.690	0.982
Friday	0.0387	0.0449	0.86	0.391	1.039
Saturday	−0.3234	0.0471	−6.87	<0.001	0.724
Sunday	−0.2943	0.0471	−6.25	<0.001	0.745

†Reference levels: slot 12:30, Monday. The weekend day type is captured by the Saturday and Sunday terms. Slots are labelled by start time.

E. Validation on unseen days

Twenty days (60 slot forecasts) were forecast one day ahead. Table IV summarises the tests and Fig. 6 the calibration. The 95% prediction intervals covered all actual counts, which the z-test does not distinguish from the nominal level ($p = 0.076$). The predicted busiest slot was correct on 60% of days against 40% by chance ($z = 1.83$, $p = 0.034$). Predicted and actual counts correlate with $r = 0.616$ ($p < 0.001$). The overload forecasts have a Brier score of 0.172 against 0.248 for the per-slot average, a skill of 31%. The daily mean absolute error was 3.93 for the model against 4.44 for the average, but the difference is not significant (paired t, $p = 0.111$; sign test 12 of 20 days, $p = 0.252$); the gain comes from weekends (3.26 against 4.81) and Fridays (4.34 against 5.84), which the average cannot tell apart from other days. The PIT values pass the Kolmogorov–Smirnov test ($p = 0.099$) but not the chi-square test ($p = 0.001$): few values fall near 0 or 1, so the forecast distributions are somewhat wider than needed, consistent with the counts being less variable than Poisson. Updating after the first slot leaves the Brier score unchanged (0.181), as expected with $\hat{v} = 0$.

Within-day carry-over is absent: Spearman correlations of observed-to-predicted ratios between consecutive slots are $\rho = 0.219$ ($p = 0.17$) and $\rho = -0.086$ ($p = 0.60$). The runs test finds no clustering of busy days (17 runs against 21.0 expected, $p = 0.20$).

VI. DISCUSSION

The per-slot Poisson model fits every slot and turns arrival counts into a direct answer: at this canteen the 15 minutes from 12:45 are overloaded on more than half of all days, and from 13:00 the pressure eases significantly. Testing the assumption slot by slot, rather than building it in as queueing studies do [3]–[5], is what makes the probabilities defensible; the pooled Pearson test adds a check that whole days do not vary more than Poisson, and here they do not.

The forecasting results are mixed and are reported as such. The day-of-week model captures the weekend drop that a per-slot average misses, which gives its overload forecasts a clear Brier skill and cuts the error on weekends by a third; it names the peak slot more often than chance, and its intervals contain the actual counts. Its overall point accuracy is not significantly better than the per-slot average over 20 test days, in line with the observation that simple fixed-effect models are hard to beat when the remaining day effects are mild [15]. With no extra day-to-day variation, the same-day update adds nothing at this canteen.

Limitations. (i) The record covers 40 days and only the first 45 minutes of lunch; later slots, exam weeks and other semesters are not observed. (ii) With 20 test days the validation has limited power. (iii) The model estimates arrival risk, not queue length, which also depends on service rates and counters. (iv) The rate is assumed constant within a slot and arrivals independent, whereas groups of friends arrive together. (v) The regression scales a whole day up or down and cannot move the peak to another slot on special days without slot-by-day interactions. (vi) Least squares on log counts stands in for Poisson regression. (vii) Capacity and risk thresholds are project decisions that must be justified for a specific canteen.

VII. CONCLUSION AND FUTURE WORK

This paper estimated the probability that canteen arrivals exceed capacity in each 15-minute slot using a Poisson model checked slot by slot, and extended it to predict a specific day's risk and peak time. On 40 recorded days the methods identified 12:45–13:00 as the high-risk slot, a significantly quieter time from 13:00, a capacity of 36 per slot needed to keep every slot below 20% risk, a weekend drop of about 26%, and an 83%

TABLE IV
HYPOTHESIS TESTS OF THE ONE-DAY-AHEAD FORECASTS (20 UNSEEN DAYS, 60 SLOT FORECASTS, $\alpha = 0.05$)

Check	Test	Statistic	p	Result	Supports the model
95% interval coverage	z-test, one proportion	z = 1.78	0.076	coverage 100.0%	Yes
Peak slot named correctly	z-test, one proportion	z = 1.83	0.034	hit rate 60.0% vs chance 40.0%	Yes
Lower daily MAE than baseline	Paired t-test	t = 1.26	0.111	MAE 3.93 vs 4.44	No
More winning days than baseline	Sign test	12 wins, 8 losses	0.252	12/20 days better	No
PIT uniform	Kolmogorov–Smirnov	D = 0.155	0.099	uniform	Yes
PIT histogram flat	χ^2 goodness of fit	χ^2 = 27.67, df = 9	0.001	not flat	No

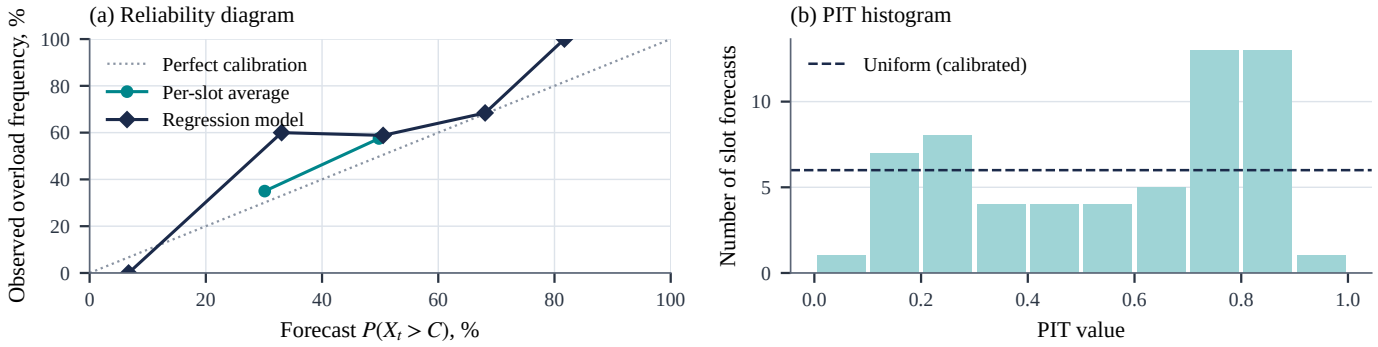


Fig. 6. Calibration on unseen days. (a) Forecast overload probability against observed overload frequency. (b) Randomised PIT values of the count forecasts; a flat histogram indicates a calibrated forecast distribution.

probability that a regular Monday's peak falls in 12:30–13:00, while showing honestly that the gain in point accuracy over a per-slot average is not significant on this record.

Future work will extend the counts to the whole lunch period and a full semester including exam weeks, add service times to estimate queue length through an M/M/c model, model group arrivals with compound Poisson distributions, estimate a smooth rate $\lambda(t)$ [9], and automate counting with turnstile or camera data.

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